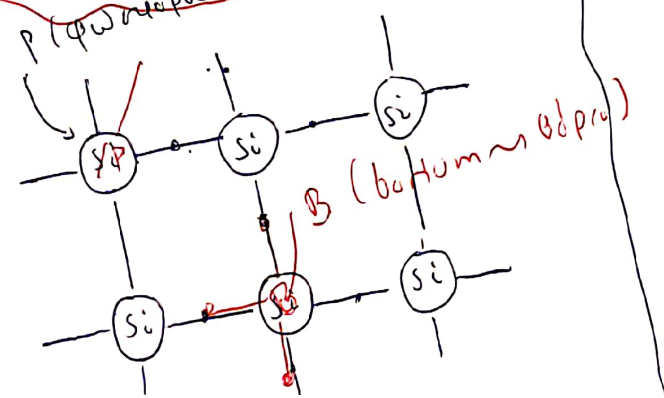
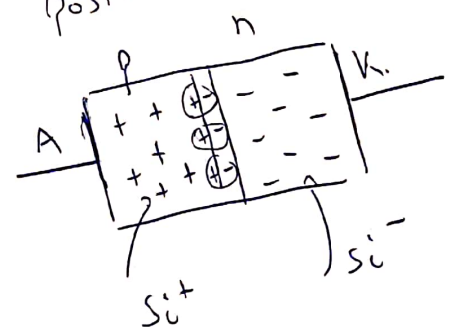


$\varphi \in \mathbb{R}$
①:

$$\varphi(x) = \frac{1}{2} V_0 \cdot \left[1 + \frac{x}{\sqrt{x^2 + a^2}} \right], (V)$$

Ερευνα $p-n$
positive
negative.



to junction with n-type
w p-doped
known to junction n-type and p-type
Boltzmann to n-type

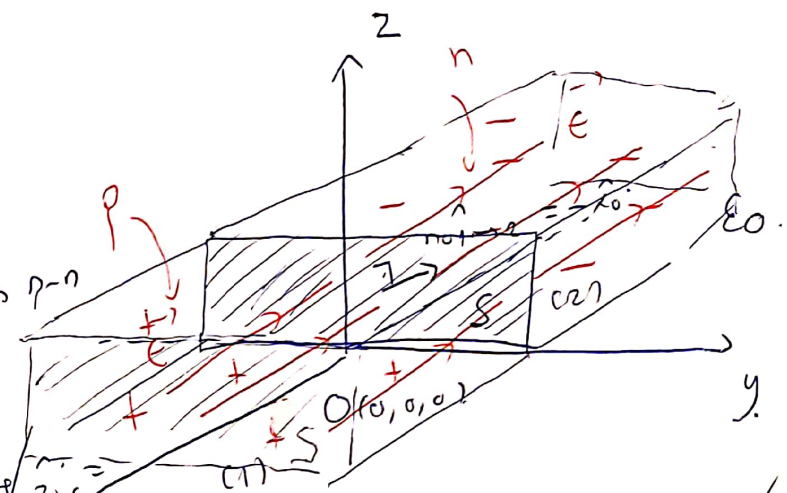
$$\vec{E} = -\nabla \varphi$$

$$= -\frac{\partial \varphi}{\partial x} \hat{x}_0 = -\frac{\partial}{\partial x} \left[\frac{1}{2} V_0 + \frac{1}{2} V_0 \frac{x}{\sqrt{x^2 + a^2}} \right] \hat{x}_0 = -\frac{1}{2} V_0 \frac{\frac{1}{\sqrt{x^2 + a^2}} - x \cdot \frac{x}{(x^2 + a^2)^{3/2}}}{1} \hat{x}_0$$

$$= -\frac{1}{2} V_0 \frac{\sqrt{x^2 + a^2} - \frac{x^2}{\sqrt{x^2 + a^2}}}{x^2 + a^2} \hat{x}_0 = -\frac{1}{2} V_0 \frac{x^2 + a^2 - x^2}{\sqrt{x^2 + a^2} (x^2 + a^2)} \hat{x}_0 = -\frac{1}{2} V_0 \frac{a^2}{(x^2 + a^2)^{3/2}} \hat{x}_0$$

$$\vec{E} = \frac{1}{2} V_0 \frac{a^2}{(x^2 + a^2)^{3/2}} (-\hat{x}_0), (V/m)$$

αφαιρέση $p \sim x > 0$, αφαιρέση $n \sim x < 0$.
Διέρρηξη $p-n \sim x = 0$.



Αρα $\int \frac{1}{(x^2 + a^2)^{3/2}} dx = \left[\frac{x}{\sqrt{x^2 + a^2}} \right] + C$

$$\vec{\nabla} \cdot \vec{D} = \rho_v = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = -\frac{1}{2} \rho_0 a^2 \frac{\partial}{\partial x} \left(\frac{1}{(x^2 + a^2)^{3/2}} \right) =$$

$$\rho_v = \frac{3}{2} v_0 \epsilon_0 a^2 \frac{x}{(x^2 + a^2)^2 \sqrt{x^2 + a^2}}$$

(C/m³)

$$-1)1 =$$

$$\left\{ \begin{array}{l} x > 0, \rho_v > 0 \\ \text{(περιεκν. ρ)} \\ x < 0, \rho_v < 0. \\ \text{(απεικν. η)} \end{array} \right.$$

$$= -\frac{1}{2} v_0 \epsilon_0 a^2 \left(-\frac{3}{2} (x^2 t_0^2)^{-3/2-1} \cdot \cancel{2x} \right) =$$

$$= \frac{3}{2} v_0 \epsilon_0 a^2 \frac{1}{(x^2 t_0^2)^{5/2}} \cdot x \quad \sim$$

$$P_s|_{x=0} = \hat{n}_{01 \rightarrow 2} (\vec{D}_2 - \vec{D}_1)|_{x=0} = \left(\frac{1}{\sqrt{2}} \right) \cdot \left[\frac{1}{2} v_0 \epsilon_0 \frac{a^2}{(a^2 + x_0^2)^{3/2}} - \frac{1}{2} v_0 \epsilon_0 \frac{a^2}{(a^2 + x_0^2)^{3/2}} \right] = \frac{1}{2} v_0 \epsilon_0 \frac{a^2}{a} - \frac{1}{2} v_0 \epsilon_0 \frac{a^2}{a} = 0 \text{ (C/m}^2\text{)}$$

$$\varphi(x) = \frac{1}{2} v_0 \left(1 + \frac{x \sqrt{m}}{\sqrt{x^2 + a_2 \sqrt{m}}} \right)$$

 $V_0 (V)$ $a(m)$

(a) $V_0, a \sim 1$

αριθμός των
σώλων = ?

(જે ૨૧ મિત્રોના
પ્રાન્ટોડ).

(e) $\in \checkmark$

(γ) p ∨ ✓

(f) $Q_0 \geq p$, $Q_0 \leq m$, $Q_0 \leq a$. ✓

$$(e) \quad \langle \omega_e \rangle$$

(or) O.I. ✓

$$(δ) \quad Q_{02\rho} = \iiint_{V\rho} \rho_v dv = \iiint_S \int_0^{+\infty} \frac{3}{2} v_0 \epsilon_0 a^2 \frac{x}{(x^2 + a^2)^{5/2}} dx dy dz = S \cdot \frac{3}{2} v_0 \epsilon_0 a^2 \left[-\frac{1}{3} \frac{1}{(x^2 + a^2)^{3/2}} \right]_0^{+\infty} = \frac{S v_0 \epsilon_0 a^2}{2} \left(-\frac{1}{\infty} + \frac{1}{(a^2)^{3/2}} \right) = \frac{S v_0 \epsilon_0}{2 a} \quad (C)$$

$$Q_{02\eta} = \iiint_{V\eta} \rho_v dv = \iiint_S \int_{-\infty}^0 \frac{3}{2} v_0 \epsilon_0 a^2 \frac{x}{(x^2 + a^2)^{5/2}} dx dy dz =$$

$\iint_S dy dz = S$ ~ αφού έχω αντίμετρα
 στο y, z και αυθαίρετο
 επιβάλλον το οποίο $\eta-\eta$

$$(ε) \quad \langle w_e \rangle = \frac{1}{2} \epsilon |\vec{E}|^2 \quad (J/m^3)$$

χωρίς συνδυασμό
 της η/σ επιρροής.

$$\vec{D} = \epsilon \vec{E}$$

$$\langle w_e \rangle = \frac{1}{2} \vec{D} \cdot \vec{E} = \frac{1}{2} |\vec{D}| |\vec{E}| \cos 0 = \frac{1}{2} \epsilon |\vec{E}| |\vec{E}| = \frac{1}{2} \epsilon |\vec{E}|^2$$

$$S \cdot \frac{3}{2} v_0 \epsilon_0 a^2 \left[-\frac{1}{3} \frac{1}{(x^2 + a^2)^{3/2}} \right]_{-\infty}^0 = \frac{S v_0 \epsilon_0 a^2}{2} \left(-\frac{1}{\infty} + \frac{1}{(a^2)^{3/2}} \right) = -\frac{3}{2} (x^2 + a^2)^{-3/2} \cdot x = -\frac{3x}{(x^2 + a^2)^{5/2}} \sim$$

$$= -\frac{S v_0 \epsilon_0}{2 a} \quad (C)$$

$$Q_{02\rho} + Q_{02\eta} = Q_{02\sigma} = 0$$

$$\sim \int \frac{x}{(x^2 + a^2)^{5/2}} = \left[-\frac{1}{3} \frac{1}{(x^2 + a^2)^{3/2}} \right]$$

$$\oint \vec{\eta} d\vec{s} = \alpha \epsilon r u \hat{z}. \quad (\Rightarrow) \quad \int_0^{2\pi} \int_0^{2\pi} D(\rho) \cancel{\rho} d\varphi dz \cancel{\rho_0} = \int_0^{2\pi} \int_0^{2\pi} \int_0^P \rho v(\rho) \rho d\rho d\varphi dz. \rightarrow$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}$$

$$\rightarrow D(\rho) \cdot \rho \cdot \int_0^{2\pi} \int_0^{2\pi} \cancel{dz} = \int_0^{2\pi} \int_0^{2\pi} \int_0^P \frac{a}{\cancel{\rho}} e^{-\rho/\epsilon} \cancel{\rho} d\rho \rightarrow$$

$$\rightarrow D(\rho) \cdot \rho = \int_0^P a e^{-\rho/\epsilon} d\rho \sim D(\rho) \cdot \rho = a \left[\frac{1}{-\frac{1}{\epsilon}} e^{-\rho/\epsilon} \right]_0^P \sim D(\rho) \cdot \rho = -a\epsilon (e^{-P/\epsilon} - 1) \sim$$

$$\sim D(\rho) \cdot \rho = -a\epsilon (e^{-P/\epsilon} - 1) \sim D(\rho) = \frac{1 - e^{-P/\epsilon}}{a\epsilon \rho} \hat{\rho}_0 \text{ (C/m}^2\text{)}$$

$$\vec{E}(\rho) = \frac{\vec{D}(\rho)}{\epsilon(\rho)} = \frac{1 - e^{-P/\epsilon}}{a\epsilon \rho (1 - e^{-P/\epsilon})} \hat{\rho}_0 \sim \vec{E}(\rho) = \frac{1}{2a\epsilon \rho^2} \hat{\rho}_0 \text{ (V/m)}$$

$$\left. \begin{aligned} E(\omega) &= D(\omega) = \\ &= \varphi(\omega) = 0 \end{aligned} \right\}$$

$$\varphi_{P \rightarrow P\omega} = \int_P^{P\omega} \vec{E} d\vec{x} = \int_P^{P\omega} \frac{1}{2a\epsilon \rho^2} \hat{\rho}_0 d\rho \cancel{\rho_0} = -\frac{1}{2a\epsilon} \left(\frac{1}{\rho} \right)_P^{P\omega} = -(2a\epsilon)^{-1} \left(\frac{1}{\cancel{\rho\omega}} - \frac{1}{P} \right) =$$

$$= \frac{1}{2a\epsilon \rho} \text{ (V)}$$

$$u(\omega) = \frac{1}{2} \epsilon(\rho) \cdot |\vec{E}(\rho)|^2 = \frac{1}{2} \rho (1 - e^{-P/\epsilon}) \frac{1}{2^2 a^2 \epsilon^2 \rho^4} \int \frac{(1 - e^{-P/\epsilon})}{2^2 a^2 \epsilon^2 \rho^3} \text{ (J/m}^3\text{)}$$

Δεν υπάρχει, γιατί $\rho(\omega) = 0$,
άρα δεν υπάρχει φορτίο στο ω

①:



-6-

(1) $0 \leq r \leq R_1$

(2) $R_1 \leq r \leq R_2 \rightsquigarrow p_v(r), Q_{00} = Q_{\cdot},$
 $\rightarrow \rightarrow (r - R_1)$
 $\rightarrow \rightarrow \rightarrow$

$$60 \text{ } 25 \times 0.3 = +12$$

(3) $r > R_2$.

$$k = \underbrace{|\vec{n}|}_{\text{wavelength}} \hat{n}_0$$

(a) $\vec{\kappa} = ?$ entspricht zu $\varepsilon, \rho_1, \rho_2, Q$

(b) $\varphi_{1,2,3} = ?$

(1) $\oint_S \vec{D}_1 \cdot d\vec{S}_1 = Q_{\text{enc}} \rightarrow \int_0^{2\pi} \int_0^\pi D_1 r^2 \sin\theta d\theta d\phi = 0 \sim D_1 = E_1 = 0$

$$(3) \oint_S \vec{D}_3 \cdot d\vec{S}_3 = Q_{enc3} \longrightarrow \int_0^{r_0} \int_0^\pi \underbrace{D_3 r \sin\theta \, r^2 \sin\theta \, d\theta \, d\phi}_{D_3 4\pi r^2} = +Q \sim$$

$$D_3 = \frac{Q}{4\pi r^2} \hat{r}_0, \text{ (C/m}^2\text{)}$$

$$\vec{E}_3 = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}_0, (V/m)$$

$$(2) \oint_S \vec{D}_2 \cdot d\vec{S}_2 = Q(r) \quad \downarrow$$

$$L_1: \int_0^{2\pi} \int_0^\pi \int_0^{R_1} D_2 r^2 \sin \theta d\theta d\varphi = \int_0^{2\pi} \int_0^\pi \int_0^{R_1} \rho_V(r) r^2 \sin \theta dr d\theta d\varphi$$

$$D_2 = \epsilon \epsilon_2 =$$

$$= \epsilon |\vec{k}| (r - R_1) \hat{r} \cdot \hat{r}$$

$$\left[\int_a^b f(x) dx \right]' = [f(x)]_a^b = f(b) - f(a)$$

$$Q = \int_0^{2\pi} \int_0^\pi \int_0^{R_1} \rho_V(r) r^2 \sin \theta dr d\theta d\varphi$$

$$Q = \int_0^{2\pi} d\varphi \int_0^\pi \sin \theta d\theta \int_0^{R_1} (3\epsilon\epsilon - \frac{2\epsilon\epsilon R_1}{r}) r^2 dr$$

$$Q = 2\pi \cdot 2 \cdot \int_0^{R_1} (3\epsilon\epsilon r^2 - 2\epsilon\epsilon R_1 \cdot r) dr$$

$$Q = 4\pi \left(\frac{3}{3} \epsilon\epsilon \left(\frac{r^3}{3} \right) \Big|_{R_1}^{R_2} - \epsilon\epsilon R_1 \left(\frac{r^2}{2} \right) \Big|_{R_1}^{R_2} \right)$$

$$\rightarrow \int_0^{2\pi} \int_0^\pi \int_0^{R_1} \epsilon k (r - R_1) r^2 \sin \theta d\theta d\varphi dr = \int_0^{2\pi} d\varphi \int_0^\pi \sin \theta d\theta \int_0^{R_1} \rho_V(r) \cdot r^2 dr$$

$$\rightarrow 4\pi \epsilon k (r - R_1) r^2 = 4\pi \int_0^r \rho_V(r) \cdot r^2 dr \rightarrow$$

$$\rightarrow \epsilon k r^3 - \epsilon k R_1 r^2 = \int_0^r \rho_V(r) r^2 dr \xrightarrow{()'} \rightarrow$$

$$\rightarrow 3\epsilon k r^2 - 2\epsilon k R_1 r = [\rho_V(r) \cdot r^2]_0^r \rightarrow$$

$$\rightarrow 3\epsilon k r^2 - 2\epsilon k R_1 \cdot r = \rho_V(r) \cdot r^2 - \rho_V(R_1) \cdot R_1^2$$

$$\rho_V(r) = 3\epsilon\epsilon - \frac{2\epsilon\epsilon R_1}{r}$$

Άρα έχω το $\rho_V(r)$ συνάρτηση του k

✓ 0
άρα λοιπόν
σφαιρικό χωρικό
φορτίο $\rho_V(r)$
από την R_1

$$\hookrightarrow Q = 4\pi \left[\epsilon (R_2^3 - R_1^3) - \epsilon \kappa R_1 (R_2^2 - R_1^2) \right] \sim Q = \kappa \left[4\pi \epsilon (R_2^3 - R_1^3) - \epsilon R_1 (R_2^2 - R_1^2) \right]$$

$$\vec{E}_2 = \frac{Q (r - R_1)}{4\pi \epsilon (R_2 - R_1) R_2^2} \hat{r}_0, (V/m)$$

$$\kappa = \frac{Q}{4\pi \epsilon [R_2^3 - R_1^3] - \epsilon R_1 [R_2^2 - R_1^2]}$$

$$\Phi_3 = \Phi_{P \rightarrow P_\omega} = \int_P^{P_\omega} \vec{E}_3 \cdot d\vec{l} = \int_r^\omega \frac{Q}{4\pi \epsilon_0 r^2} \hat{r}_0 \cdot d\vec{r} = \frac{Q}{4\pi \epsilon_0} \frac{1}{r}, (V) = \frac{Q}{4\pi \epsilon R_2^3 - 4\pi \epsilon R_1^3 - 4\pi \epsilon R_1 R_2^2 + 4\pi \epsilon R_1^3}$$

$$\Phi_2 = \Phi_{M \rightarrow P_\omega} = \int_M^K \vec{E}_2 \cdot d\vec{l} + \underbrace{\int_{K_1}^{P_\omega} \vec{E}_3 \cdot d\vec{l}}_{\Phi_3(R_2)} = \int_r^{R_2} \frac{Q (r - R_1)}{4\pi \epsilon (R_2 - R_1) R_2^2} \hat{r}_0 \cdot d\vec{r} + \Phi_3(R_2) \sim \kappa = \frac{Q}{4\pi \epsilon [R_2 - R_1] R_2^2}$$

$$= \frac{Q}{4\pi \epsilon (R_2 - R_1) R_2^2} \left[\frac{R_2^2 - r^2}{2} - R_1 (R_2 - r) \right] + \frac{Q}{4\pi \epsilon_0 R_2}$$

$$\Phi_1 = \Phi_{\pm \rightarrow P_\omega} = \int_{\pm}^{\omega} \vec{E}_1 \cdot d\vec{l} + \underbrace{\int_{\pm}^K \vec{E}_2 \cdot d\vec{l}}_{\Phi_2(R_1)} + \int_K^{P_\omega} \vec{E}_3 \cdot d\vec{l} = \Phi_2(R_1), \text{ οπότε, τους σφαιρικούς τους μεσαί φαιράς ισοδυναμεί } \underline{\text{μεσαί φαιράς ισοδυναμεί}}$$